

Sai Venkata

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of experience designing, building, and optimizing cloud-scale data platforms and real-time data pipelines across Azure, AWS, and GCP (POC level) environments. Proven expertise in ETL/ELT pipeline development, Lakehouse architectures, real-time streaming systems, and data warehousing solutions supporting enterprise analytics and machine learning workloads.

Specialized in Azure Data Factory, Databricks, Synapse Analytics, Spark (PySpark/Scala), Kafka, and Snowflake, delivering high-performance distributed data solutions. Strong command of SQL optimization, big data processing, workflow orchestration, and CI/CD automation. Experienced in implementing data governance, security, and compliance frameworks for healthcare and banking domains (HIPAA, GDPR, PCI DSS).

Recognized for translating complex business requirements into scalable, production-grade data solutions, improving performance, reliability, and time-to-insight.

CORE TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Scala, Java, Shell

Big Data & Processing: Apache Spark (PySpark, Scala), Hadoop, Hive, Kafka, Databricks, Delta Lake

Cloud Platforms:

Azure: ADLS Gen2, Data Factory, Databricks, Synapse Analytics, Azure SQL, Functions, Event Hubs, Key Vault, Purview.

AWS: S3, EC2, Lambda, Glue, Athena, Redshift.

GCP (POC): BigQuery, Dataflow.

ETL & Orchestration: Azure Data Factory, Apache Airflow, AWS Glue, Informatica.

Databases: SQL Server, PostgreSQL, Oracle, MySQL, MongoDB, Cassandra, Cosmos DB.

Data Warehousing: Snowflake, Azure Synapse, Amazon Redshift.

Streaming: Kafka, Spark Structured Streaming, Azure Event Hubs, AWS Kinesis.

BI & Analytics: Power BI, Tableau.

DevOps & CI/CD: Azure DevOps, GitHub, Jenkins, Docker, Kubernetes.

Monitoring: Azure Monitor, Log Analytics, Splunk.

Security & Governance: Azure Key Vault, Purview, IAM, Row-Level Security, Encryption.

Methodologies: Agile (Scrum/Kanban), Jira.

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Aug 2024 to Present

CVS Health, Texas

Responsibilities:

- Designed and developed end-to-end ETL pipelines using Azure Databricks (PySpark) and Azure Data Factory, ingesting patient data from 30+ EHR systems into ADLS Gen2.
- Architected a Lakehouse (Bronze–Silver–Gold / Medallion) data model, improving downstream analytics performance and ML model training efficiency by 40%.
- Led on-prem PostgreSQL to Azure migration, converting datasets into Parquet/Delta format, reducing storage costs and query latency.
- Implemented real-time streaming pipelines using Azure Event Hubs, Kafka, and Spark Structured Streaming, enabling near real-time monitoring of patient vitals.
- Optimized Spark transformations and cluster configurations, reducing pipeline runtimes by 50% and lowering compute costs.
- Built reusable ingestion frameworks supporting JSON, CSV, TXT, AVRO, improving onboarding time for new data sources by 60%.
- Automated workflow orchestration using Apache Airflow and ADF, ensuring SLA adherence and pipeline reliability.

- Implemented data quality validation frameworks in Databricks, increasing clinical data accuracy and consistency.
- Developed Power BI dashboards delivering executive-level healthcare KPIs, reducing reporting turnaround time by 40%.
- Enforced HIPAA-compliant security architecture using Azure Key Vault, Managed Identities, encryption at rest/in transit, and audit logging.
- Established CI/CD pipelines in Azure DevOps for Databricks notebooks and ADF deployments across environments.
- Monitored distributed workloads using Azure Monitor and Log Analytics, enabling proactive incident resolution.
- Collaborated in Agile teams, contributing to sprint planning, peer reviews, and mentoring junior engineers.

Data Engineer

Oct 2022 to Jul 2024

Bank of America, Plano, TX

Responsibilities:

- Built high-volume ETL pipelines using Azure Databricks (Spark-Scala) and Azure Data Factory, processing 5+ TB of daily banking transactions.
- Implemented real-time ingestion pipelines with Kafka, Azure Event Hubs, and Stream Analytics, reducing fraud detection latency by 40%.
- Migrated SQL Server data to Azure Blob Storage and Delta Lake, improving query performance and reducing infrastructure cost.
- Developed multi-format ingestion pipelines (JSON, CSV, AVRO) into curated Delta tables supporting risk and compliance analytics.
- Optimized Spark transformations, decreasing ETL execution time by 30%.
- Designed Airflow DAGs orchestrating batch settlement, loan disbursement, and transaction monitoring workflows.
- Implemented Great Expectations-based data validation, improving data integrity for AML and KYC datasets.
- Delivered consumption-ready datasets to Azure Synapse Analytics for regulatory and treasury reporting.
- Integrated Azure Purview for data classification and lineage; enforced row-level security for sensitive financial data.
- Secured credentials and encryption keys using Azure Key Vault, protecting PII data across pipelines.
- Built Power BI fraud monitoring dashboards, improving fraud detection efficiency by 35%.
- Maintained version control and CI/CD using GitHub and Azure DevOps.
- Actively participated in Agile sprints and mentored junior team members.

Data Engineer

May 2020 to Oct 2022

Mastercard, Purchase, NY

Responsibilities:

- Developed batch and real-time ingestion pipelines using Azure Databricks (Spark-Scala) and Azure Data Factory, handling millions of daily records.
- Ingested structured and semi-structured data into ADLS Gen2 and Delta Lake, enabling scalable enterprise analytics.
- Migrated legacy Oracle and SQL Server systems to Azure SQL Database, improving query response time by 35%.
- Built event-driven pipelines using Kafka and Azure Event Hubs for real-time monitoring use cases.
- Integrated Cosmos DB for NoSQL document storage and application-level analytics.
- Developed Power BI and Tableau dashboards supporting leadership reporting.
- Automated workflows using Azure Logic Apps and Apache Airflow, improving operational efficiency.
- Implemented data validation checks within ADF pipelines, increasing data quality and reducing manual reconciliation.
- Enforced Row-Level Security across Synapse and Power BI datasets.
- Configured Azure Key Vault for secure secrets management.
- Monitored Spark workloads using Azure Monitor and Log Analytics.
- Collaborated with cross-functional Agile teams to deliver on-time project milestones.

EDUCATION:

Masters in Computer Science from University of Central Missouri.